Eric Nai-Li Chen

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EDUCATION

Cornell Tech Aug 2026 - Present
Ph.D. in Information Science
Advisors: Max Kreminski, Thijs Roumen

Brown University Sep 2023 – May 2025
M.S. in Computer Science
Advisor: Jeff Huang

UCLA Sep 2018 – Jun 2023
B.S. in Computer Science & Minor in Mathematics

RESEARCH EXPERIENCE

Cornell Tech, Ph.D. Student — New York, NY
Advisors: Max Kreminski, Thijs Roumen — Aug 2026 – Present

- Researching expressive interaction techniques to support human creativity

BU Interactive Computing Lab, Visiting Researcher — Boston, MA
Mentor: Chang Xiao — Jan 2026 – July 2026

- Developed a system reframing formal grammar as a fluid design material
- Studied how 8 designers using the tool shared and repurposed other's rules

Adobe Research, Research Intern — San Jose, CA
Mentors: Li-Yi Wei, Rubaiat Habib Kazi — Jun 2025 – Sep 2025

- Designed generative palette interface for authoring elemental effect animations
- Implemented semantic sliders, linkable to stroke dynamics

Brown University HCI Lab, Research Assistant — Providence, RI
Advisor: Jeff Huang — Feb 2024 – May 2025

- Developed L-system-based drawing tool enabling strokes that grow into organic branching structures
- Investigated how the tool's combination of control & surprise impacted 12 artists

NASA Academic Mission Services (USRA), Data Science Intern — Mountain View, CA
Mentors: David Bell, Aditya Das, Milad Memarzadeh — Jun 2020 – Sep 2021

- Led multi-organizational machine learning project on wildfire prevention

PUBLICATIONS

Conference Papers

L.ink: Procedural Ink Growth for Controllable Surprise
UIST 2025
Eric Nai-Li Chen, Joshua Kong Yang, Jeff Huang, Tongyu Zhou

Extended Abstracts & Pictorials

SVGraffiti: Remixing the Web with Vector Illustrations

DIS Pictorials 2026
Tongyu Zhou, Joshua Kong Yang, *Eric Nai-Li Chen*, Jeff Huang

Actionbrushes: Painting with Elemental Dynamics from a Generative Palette
CHI EA 2026
Eric Nai-Li Chen, Rubaiat Habib Kazi, Li-Yi Wei

L.ink: Illustrating Controllable Surprise with L-System Based Strokes
CHI EA 2025
Eric Nai-Li Chen, Tongyu Zhou, Joshua Kong Yang, Jeff Huang

INVITED TALKS

Continuous Creative Channels Between Artist and Medium

Guest Lecture for CS 598X1: HCI & HAI at Boston University (host: Prof. Chang Xiao) — Mar 24, 2026

UCSD Design Lab Seminar (host: Zhicheng Huang) — Oct 22, 2025

ACADEMIC SERVICE

Conference Reviewer

2027: CHI (x4)
2026: CHI Posters, ACM C&C, ACM DIS (x2)
2025: CHI Late-Breaking Work

SELECTED PROJECTS

SplatBrush: XR Painting with 3D Gaussians — CSCI 2951-I: Computer Vision for Graphics and Interaction

- Designed WebXR interface for interactive 3D painting using Gaussian textures optimized from real-world materials

Differentiable Rendering of Signed Distance Fields — CSCI 2240: Advanced Computer Graphics

- Implemented the method of Wang et al., *A Simple Approach to Differentiable Rendering of SDFs*

Video Analogies — CSCI 1290: Computational Photography

- Implemented patch-based style transfer from images to videos, extending Hertzmann et al., *Image Analogies*

SKILLS

AI: LLM/VLM integration and application development, real-time AI inference, deep learning in graphics and vision

Technical: JavaScript/TypeScript, HTML/CSS, C/C++, Python, React, Node.js, OpenGL, TensorFlow

Methodologies: Rapid prototyping, empirical user studies, mixed-methods research, creativity support tool evaluation